

HOMEWORK 7

Problem 1

Given the provided dataset “HW7_data.csv”, consider the non-linear regression model,

$$y_i = \exp(\theta x_i) + \sigma \epsilon_i, \quad i = 1, 2, \dots, n,$$

where $n = 100$.

- (1) (15 pts) Use a Newton–Raphson algorithm to find the least squares estimator, $\hat{\theta}$, for θ (take the initial value $\theta_0 = 1$). Show all your working and algorithm.
- (2) (5 pts) What is your result for $\hat{\theta}$?
- (3) (15 pts) Use the parametric Bootstrap to find an approximate variance for $\hat{\theta}$. Show all your working and algorithm.
- (4) (5 pts) What is your result for $\text{Var}(\hat{\theta})$?
- (5) (20 pts) The logarithm of the predicted value at $x = \bar{x}$ is given by

$$\log \hat{y} = \hat{\theta} \bar{x}.$$

Find a 95% confidence interval for $\log \hat{y}$.

Problem 2

Consider the general non-linear model with σ known;

$$y_i = m(\theta, x_i) + \sigma \epsilon_i.$$

The $\hat{\theta}$ minimizes

$$l(\theta) = \frac{1}{2} \sum_{i=1}^n (y_i - m(\theta, x_i))^2,$$

and write θ_0 to be the true value.

- (1) (5 pts) Find $E l'(\theta_0)$ and $E l''(\theta_0)$, leaving your answers in terms of the unknown θ_0 .
- (2) (5 pts) Prove that $\sigma^2 E l''(\theta_0) = \text{Var } l'(\theta_0)$.
- (3) (10 pts) Using the approximation

$$l'(\theta_0) \approx l'(\hat{\theta}) + (\theta_0 - \hat{\theta}) l''(\hat{\theta}),$$

which is a first order Taylor expansion, show why we can write

$$\hat{\theta} \approx \theta_0 - \frac{l'(\theta_0)}{C},$$

where $C = l''(\hat{\theta})$ and is taken to be fixed.

- (4) (10 pts) Now using the results in part (1) to (3), what would a normal approximation of $\hat{\theta}$ be?
- (5) (10 pts) Hence, construct an approximate $(1 - \alpha)\%$ confidence interval for θ_0 .